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1. Scope and Objectives

This engine evaluates open-ended Indian **equity and hybrid mutual funds** (direct or regular plans).

Primary objectives:

- Provide a **DecisionScoreMF $\in$ ** per fund representing the strength of the investment case.
- Generate an explicit **BUY / HOLD / TRIM / SELL** recommendation.
- Decompose this into interpretable layers:
  - Fundamental Quality
  - Manager Skill

- Forward Signals
- Regime Robustness
- Portfolio Intelligence
- Edge (USP)
- Risk & Stress Behavior
- Time-Structure (1Y / 3Y / 5Y)
- Cost-Efficiency

The design is:

- **Causal-oriented**: focuses on manager alpha and portfolio behavior, not just raw returns.
- **Temporal**: uses rolling windows, trends, and stability.
- **Regime-aware**: uses macro regimes (bull, bear, high-vol, rate regimes).
- **ML-ready**: exposes both raw features and normalized scores.

***

2. Notation and Core Objects

Evaluation date: T.

For each fund f:

- $NAV_{f,t}$: daily NAV at trading day t.
- $R_{f,t} = \dfrac{NAV_{f,t} - NAV_{f,t-1}}{NAV_{f,t-1}}$: daily simple return.
- Benchmark index level $I_{c,t}$ for category/benchmark c.
- $R_{B(f),t} = \dfrac{I_{c,t} - I_{c,t-1}}{I_{c,t-1}}$: daily benchmark return.
- Risk-free rate $r^{(rf)}_t$ (e.g., T-Bill).

Windows (trading days):

- 1Y: $W_{1Y} = 252$
- 3Y: $W_{3Y} = 756$
- 5Y: $W_{5Y} = 1260$ (if available)

Window total return for fund:

$$
Ret_{f,W}(T) = \prod_{k=0}^{W-1} (1 + R_{f,T-k}) - 1
$$

Similarly $Ret_{B(f),W}(T)$ for benchmark.

Peer sets:

- Primary peer set for fund f: all funds in same SEBI category.
- If category peer count $N_{cat} < 15$: fallback peer set = all funds in same broad asset class (Equity, Hybrid).

***

3. Data Requirements and Pre-Processing

For each fund f and benchmark:

1. NAV & index prices
   - Daily NAV $NAV_{f,t}$ for at least 5 years where available.
   - Daily benchmark index $I_{c,t}$ for same window.

2. Portfolio snapshots
   - Holdings $w_{f,h}(q)$ per holding h, at least quarterly for 3 years.
   - Sector weights per quarter.
   - Style factors per stock (e.g., size and value scores).

3. Flows and AUM
   - Monthly AUM $AUM_{f,m}$ for last 36 months.
   - Monthly net subscription/redemption if separately available; otherwise approximated from AUM and NAV moves.

4. Fund metadata
   - Category_f (SEBI category).
   - PlanType_f ∈ {DIRECT, REGULAR}.
   - TaxCategory_f ∈ {Equity, NonEquity}.
   - ExpenseRatio_f (plan-specific).
   - ExitLoad_f (typical horizon).

- FundAge_f (years since launch).
- ManagerTenure_f (years current lead manager in charge).

5. Macro regime labels (external)
Daily or monthly flags:
- Bull / Bear
- HighVol
- RisingRate
- FallingRate

6. Crash windows
- Predefined crash date ranges (e.g., March 2020, 2022 correction) with weights.

Pre-processing:

- Align dates and fill holidays/holidays with previous NAV to maintain daily sequence.
- Ensure all daily and monthly series are continuous (no gaps for the engine).

***

4. Normalization and Missing-Data Rules

4.1 Percentile normalization

For any scalar metric X_f over peer set $\mathcal{P}$ of size N:

- If all values equal: $Score(X_f) = 0.5$.
- Else:
  - Sort ascending; $Rank_f \in \{1,\dots,N\}$ where 1 = worst, N = best.
  - Score:

$$
Score(X_f) = \frac{Rank_f - 1}{N - 1}
$$

For **lower-is-better** metric Y_f:

$$
Score(Y_f) = Score(-Y_f)
$$

\]

All composite scores are averages of such primitive scores with non-negative weights, hence guaranteed to be in \([0,1]\).

4.2 Missing data

If primitive metric for a fund is missing:

- Use neutral score 0.5 in places where that metric is included.
- Expose a DataCoverage flag separately in ML layer.

***

5. LAYER 1 – FUNDAMENTAL QUALITY

Goal: measure realized, risk-adjusted performance quality with consistency, without redundancy.

5.1 Inputs

For each fund \(f\):

- Daily returns \(R_{f,t}\), benchmark returns \(R_{B(f),t}\), risk-free rate \(r^{(rf)}_t\) for last 3 years (mandatory) and 5 years (optional).

5.2 Risk-adjusted returns (3Y horizon, primary)

Compute over last \(W_{3Y}\) days:

- Average daily excess over risk-free:

\[
\bar{r}^{(rf)}_{f,3Y} = \frac{1}{W_{3Y}}\sum_{k=0}^{W_{3Y}-1}(R_{f,T-k} - r^{(rf)}_{T-k})
\]

- Volatility:

\[
\sigma_{f,3Y} = SD(R_{f,t})_{t \in [T-W_{3Y}+1,T]}

\]

Sharpe:

\[
Sharpe_{f,3Y} = \frac{\bar{r}^{(rf)}_{f,3Y}}{\sigma_{f,3Y}} \sqrt{252}
\]

Sortino:

- Downside deviations:

\[
DD_{f,t} = \min(0,\ R_{f,t} - r^{(rf)}_t)
\]

\[
\sigma^{-}_{f,3Y} = SD(DD_{f,t})_{t \in [T-W_{3Y}+1,T]}
\]

\[
Sortino_{f,3Y} = \frac{\bar{r}^{(rf)}_{f,3Y}}{\sigma^{-}_{f,3Y}} \sqrt{252}
\]

5.3 Drawdown stability

Compute maximum drawdown over last 3 years:

Let cumulative path $C_t = NAV_{f,t}$ over that range.

\[
Peak_t = \max_{u \le t} C_u,\quad DD_t = \frac{C_t - Peak_t}{Peak_t}
\]

\[
MaxDD_{f,3Y} = \min_{t} DD_t
\]

5.4 Consistency – hit ratio

Define monthly 1Y rolling windows over last 3Y; for each month m:

- Fund 1Y return: $Ret_{f,1Y}(m)$.
- Benchmark 1Y: $Ret_{B,1Y}(m)$.
- Excess: $Excess1Y_f(m) = Ret_{f,1Y}(m) - Ret_{B,1Y}(m)$.

Hit ratio:

$$
HitRatio_{f,3Y} = \frac{\#\{m: Excess1Y_f(m) > 0\}}{K}
$$

5.5 Scores

- $SharpeScore_f = Score(Sharpe_{f,3Y})$
- $SortinoScore_f = Score(Sortino_{f,3Y})$
- $DDScore_f = Score(-MaxDD_{f,3Y})$
- $HitScore_f = Score(HitRatio_{f,3Y})$

Risk-adjusted performance:

$$
\begin{aligned}
RAPScore_f = {} & 0.35 \cdot SharpeScore_f \\
& + 0.35 \cdot SortinoScore_f \\
& + 0.30 \cdot HitScore_f
\end{aligned}
$$

Drawdown stability:

$$
DrawdownStability_f = DDScore_f
$$

5.6 FundamentalQualityScore

$$
\begin{aligned}
FundamentalQuality_f = {} & \\
& 0.75 \cdot RAPScore_f
\end{aligned}
$$

$$
+0.25 \cdot DrawdownStability_f
$$

Interpretation: realized risk-adjusted performance with drawdown behavior.

***

6. LAYER 2 – MANAGER SKILL

Goal: treat manager as alpha generator vs benchmark.

6.1 Inputs

- Daily returns for last 3 years (min) for fund and benchmark.

6.2 Rolling alpha and IR (12-month windows)

For each rolling 12-month window ending at month m:

Regression:

$$
R_{f,t} = \alpha_{f,m} + \beta_{f,m} R_{B(f),t} + \epsilon_t
$$

Alpha (annualized):

$$
Alpha_{f,m} = 252 \cdot \alpha_{f,m}
$$

Residual volatility:

$$
\sigma_{\epsilon,f,m} = SD(\epsilon_t)
$$

Information ratio:

$$
IR_{f,m} = \frac{\alpha_{f,m}}{\sigma_{\epsilon,f,m}} \sqrt{252}
$$

6.3 Aggregate alpha metrics

Over all K windows in last 3Y:

$$
AlphaMean_f = \frac{1}{K}\sum_{m=1}^{K} Alpha_{f,m}
$$

$$
AlphaVol_f = SD(Alpha_{f,m})_{m=1..K}
$$

$$
AlphaHit_f = \frac{\#\{m: Alpha_{f,m} > 0\}}{K}
$$

$$
IRMean_f = \frac{1}{K}\sum_{m=1}^{K} IR_{f,m}
$$

6.4 Capture ratios

Upside capture:

$$
UpsideCapture_f = \frac{\sum_{t: R_{B(f),t}>0} R_{f,t}}{\sum_{t: R_{B(f),t}>0} R_{B(f),t}}
$$

Downside capture:

$$
DownsideCapture_f = \frac{\sum_{t: R_{B(f),t}<0} R_{f,t}}{\sum_{t: R_{B(f),t}<0} R_{B(f),t}}
$$

6.5 Skill trend / decay

Fit linear trend of alpha over months:

$$
Alpha_{f,m} = a_f + b_f \cdot m + \eta_m
$$

Slope:

$$
AlphaTrend_f = b_f
$$

Decay flag:

$$
SkillDecayFlag_f =
\begin{cases}
1, & \text{if } AlphaTrend_f < 0 \text{ and } |AlphaTrend_f| > \theta_{trend} \\
0, & \text{otherwise}
\end{cases}
$$

6.6 Scores

Normalize over peers:

- $AlphaScore_f = Score(AlphaMean_f)$
- $AlphaStabilityScore_f = Score(-AlphaVol_f)$
- $AlphaHitScore_f = Score(AlphaHit_f)$
- $IRScore_f = Score(IRMean_f)$
- $UpsideScore_f = Score(UpsideCapture_f)$
- $DownsideScore_f = Score(-DownsideCapture_f)$
- $AlphaTrendScore_f = Score(AlphaTrend_f)$

Decay penalty:

$$
SkillDecayPenalty_f =

\begin{cases}
0.2, & \text{if } SkillDecayFlag_f = 1 \\
0, & \text{otherwise}
\end{cases}
\]

ManagerSkillRaw:

\[
ManagerSkillRaw_f =
0.25 \cdot AlphaScore_f
+0.15 \cdot AlphaStabilityScore_f
+0.15 \cdot AlphaHitScore_f
+0.15 \cdot IRScore_f
+0.15 \cdot UpsideScore_f
+0.15 \cdot DownsideScore_f
\]

Final:

\[
ManagerSkill_f = \max\big(0,\ ManagerSkillRaw_f - SkillDecayPenalty_f\big)
\]

***

7. LAYER 3 – FORWARD SIGNAL LAYER

Goal: proxy for *future* outperformance.

7.1 Inputs

- Alpha series \(\{Alpha_{f,m}\}\).
- Hit ratio series \(\{HitRatio_{f,1Y}(m)\}\).
- Monthly AUM and returns (for flows).
- Category average returns.

7.2 Short-term alpha trend

Use last K_{short} windows (e.g., 12 months):

$$
Alpha_{f,m} = a^{(short)}_f + b^{(short)}_f \cdot m + \eta_m
$$

$$
ForwardAlphaTrend_f = b^{(short)}_f
$$

$$
ForwardAlphaTrendScore_f = Score(ForwardAlphaTrend_f)
$$

7.3 Hit ratio change

Split 3Y alpha windows into early and late halves:

$$
HitChange_f = HitRatio_{f,new} - HitRatio_{f,old}
$$

$$
HitChangeScore_f = Score(HitChange_f)
$$

7.4 Flow momentum

Monthly flow rate:

$$
FlowRaw_{f,m} = AUM_{f,m} - AUM_{f,m-1}(1 + Ret_{f,1M}(m))
$$

$$
FlowRate_{f,m} = \frac{FlowRaw_{f,m}}{AUM_{f,m-1}}
$$

Average last 6 months:

$$
AvgFlow6_f = \frac{1}{6}\sum_{m=T-5}^{T} FlowRate_{f,m}
$$

$$
FlowMomentumScore_f = Score(AvgFlow6_f)
$$

(If you want to penalize crowding, you may later invert sign if flows are too high.)

7.5 Category-relative momentum

Category average 1Y return:

$$
CatAvgRet_{1Y} = \frac{1}{|\mathcal{P}|}\sum_{g \in \mathcal{P}} Ret_{g,1Y}
$$

$$
CatRelRet_{f,1Y} = Ret_{f,1Y} - CatAvgRet_{1Y}
$$

$$
CatRelScore_f = Score(CatRelRet_{f,1Y})
$$

7.6 ForwardOutperformanceProbability

$$
\begin{aligned}
ForwardSignalRaw_f = \\
&0.35 \cdot ForwardAlphaTrendScore_f \\
&+0.25 \cdot HitChangeScore_f \\
&+0.20 \cdot FlowMomentumScore_f \\
&+0.20 \cdot CatRelScore_f
\end{aligned}
$$

$$
ForwardOutperformanceProb_f = ForwardSignalRaw_f
$$

Interpretation: estimated probability fund will outperform over next 1–3 years.

***

8. LAYER 4 – REGIME INTELLIGENCE

Goal: regime-aware robustness.

8.1 Regimes

External labels:

- BULL, BEAR, HIGHVOL, RISINGRATE, FALLINGRATE (daily or monthly flags).

8.2 Regime performance

For regime r, let $\mathcal{T}_r$ be all days in last 5Y with $Regime_r(t)=1$.

If $|\mathcal{T}_r|$ < threshold (e.g., 60 days) → skip / neutral.

Otherwise:

$$
\bar{R}_{f,r} = \frac{1}{|\mathcal{T}_r|}\sum_{t \in \mathcal{T}_r} R_{f,t}
$$

$$
Ret_{f,r} = (1 + \bar{R}_{f,r})^{252} - 1
$$

Benchmark:

$$
\bar{R}_{B,r},\ Ret_{B,r}
$$

Excess:

$$
ExcessRet_{f,r} = Ret_{f,r} - Ret_{B,r}
$$

Sharpe in regime:

$$
Sharpe_{f,r} = \frac{\bar{R}_{f,r} - \bar{r}^{(rf)}_r}{SD(R_{f,t})_{t \in \mathcal{T}_r}} \sqrt{252}
$$

Scores:

$$
ScoreExcess_{f,r} = Score(ExcessRet_{f,r}),\quad ScoreSharpe_{f,r} = Score(Sharpe_{f,r})
$$

$$
RegimeScore_{f,r} = 0.6 \cdot ScoreExcess_{f,r} + 0.4 \cdot ScoreSharpe_{f,r}
$$

Aggregate with weights w_r:

$$
RegimeRobustness_f = \sum_r w_r RegimeScore_{f,r}
$$

***

9. LAYER 5 – PORTFOLIO INTELLIGENCE

Goal: style, diversification, and behavior of holdings.

9.1 Active Share

$$
ActiveShare_f = \frac{1}{2} \sum_h | w_{f,h} - w_{B,h} |
$$

$$
\[
$$

ActiveShareScore_f = Score(ActiveShare_f)
\]

9.2 Style exposure & consistency

Using style factors:

\[
SizeExposure_f(q) = \sum_h w_{f,h}(q) \cdot Size_h
\]
\[
ValueExposure_f(q) = \sum_h w_{f,h}(q) \cdot Value_h
\]

Across quarters \(q=1..Q\):

\[
SizeVar_f = SD(SizeExposure_f(q)),\quad ValueVar_f = SD(ValueExposure_f(q))
\]

\[
StyleConsistencyScore_f = 0.5 \cdot Score(-SizeVar_f)
                + 0.5 \cdot Score(-ValueVar_f)
\]

9.3 Sector drift

\[
SectorDrift_f = \frac{1}{Q-1}\sum_{q=2}^{Q} \Big(\frac{1}{S}\sum_s |SectorWeight_f,s(q) - SectorWeight_f,s(q-1)|\Big)
\]
\[
SectorDriftScore_f = Score(-SectorDrift_f)
\]

9.4 Turnover quality (if available)

$$
TurnoverAvg_f = \text{mean}(Turnover_f(q))
$$

$$
TurnoverScore_f = Score(-TurnoverAvg_f)
$$

9.5 PortfolioQualityScore

$$
PortfolioQualityScore_f =
 0.35 \cdot ActiveShareScore_f
+0.25 \cdot StyleConsistencyScore_f
+0.25 \cdot SectorDriftScore_f
+0.15 \cdot TurnoverScore_f
$$

***

10. LAYER 6 – EDGE / DIFFERENTIATION

Goal: MF equivalent of USP – non-consensus, value-adding deviation.

10.1 Non-consensus positioning

Category average weight for each holding:

$$
w_{Cat,h} = \frac{1}{|\mathcal{P}|}\sum_{g \in \mathcal{P}} w_{g,h}
$$

$$
NCPos_f = \frac{1}{2}\sum_h |w_{f,h} - w_{Cat,h}|
$$

$$
NCScore_f = Score(NCPos_f)
$$

10.2 Strategy uniqueness

Let style vector S_f include size, value, and sector weights. Category average S_{Cat}.

$$
StrategyDev_f = ||S_f - S_{Cat}||_2
$$

$$
StrategyDevScore_f = Score(StrategyDev_f)
$$

10.3 Active deviation with alpha

$$
ActiveWithAlpha_f = ActiveShare_f \cdot \max(0, AlphaMean_f)
$$

$$
ActiveWithAlphaScore_f = Score(ActiveWithAlpha_f)
$$

10.4 EdgeScore

$$
EdgeScore_f =
 0.40 \cdot ActiveWithAlphaScore_f
+0.30 \cdot NCScore_f
+0.30 \cdot StrategyDevScore_f
$$

***

11. LAYER 7 – RISK & STRESS BEHAVIOR

Goal: tail risk, crash behavior, recovery, and clustering.

11.1 Tail risk (CVaR)

Over last 3Y daily returns, sort $R_{f,t}$ ascending. For level $q = 0.95$:

- VaR:

$$
VaR_f^{(q)} = q\text{-quantile of }R_{f,t}
$$

- CVaR:

$$
CVaR_f^{(q)} = \text{mean of }R_{f,t} \text{ where }R_{f,t} \le VaR_f^{(q)}
$$

$$
CVaRScore_f = Score(-CVaR_f^{(q)})
$$

11.2 Crash performance

For each crash window w:

$$
CrashRet_{f,w} = \prod_{t \in w}(1 + R_{f,t}) - 1
$$

$$
CrashRet_{B,w} = \prod_{t \in w}(1 + R_{B(f),t}) - 1
$$

$$
CrashExcess_{f,w} = CrashRet_{f,w} - CrashRet_{B,w}
$$

Aggregate with weights ω_w:

$$
CrashExcess_f = \sum_w \omega_w CrashExcess_{f,w}
$$

$$
\[
$$

CrashScore_f = Score(CrashExcess_f)
\]

11.3 Recovery time

For each major drawdown event \(j\):

- Identify peak, trough, and recovery back to peak.
- RecoveryDays_f^{(j)} = time to recover.

Average:

\[
RecoveryAvg_f = \text{mean}(RecoveryDays_f^{(j)})
\]
\[
RecoveryScore_f = Score(-RecoveryAvg_f)
\]

11.4 Volatility clustering

Let \(X_t = R_{f,t}^2\). Lag-1 autocorrelation:

\[
VolCluster_f = Corr(X_t, X_{t-1})
\]
\[
VolClusterScore_f = Score(-VolCluster_f)
\]

11.5 RiskBehaviorScore

\[
RiskBehaviorScore_f =
0.35 \cdot CVaRScore_f
+0.25 \cdot CrashScore_f
+0.25 \cdot RecoveryScore_f

+0.15 \cdot VolClusterScore_f
\]

***

12. LAYER 8 – TIME-STRUCTURE

Goal: use 1Y / 3Y / 5Y structure, capture trend + stability.

12.1 Horizon-specific scores

For each horizon \(H ∈ \{1Y,3Y,5Y\}\) (if data):

- Compute FundamentalQuality_{f,H}, ManagerSkill_{f,H}, RiskBehavior_{f,H} with same formulas using window \(W_H\).

12.2 Trends

For metric \(M \in \{\text{FundamentalQuality},\ \text{ManagerSkill},\ \text{RiskBehavior}\}\):

Define \(M_{f,1Y}, M_{f,3Y}, M_{f,5Y}\).

Trend:

\[
Trend_M_f = \frac{M_{f,5Y} - M_{f,1Y}}{5 - 1}
\]
\[
TrendScore_M_f = Score(Trend_M_f)
\]

12.3 Stability

\[
Var_M_f = Var(M_{f,1Y}, M_{f,3Y}, M_{f,5Y})
\]
\[
StabilityScore_M_f = Score(-Var_M_f)
\]

12.4 TimeStructureScore

$$
TimeStructureScore_f =
\frac{1}{3}\sum_M \big(0.6 \cdot TrendScore_M_f + 0.4 \cdot StabilityScore_M_f\big)
$$

***

13. LAYER 9 – COST-EFFICIENCY

Goal: cost relative to alpha and net excess.

13.1 Alpha per cost

$$
AlphaPerCost_f = \frac{AlphaMean_f}{ExpenseRatio_f + \epsilon}
$$

$$
AlphaPerCostScore_f = Score(AlphaPerCost_f)
$$

13.2 Net alpha after cost

$$
ExcessRet_{f,3Y} = Ret_{f,3Y} - Ret_{B(f),3Y}
$$

$$
NetExcessRet_{f,3Y} = ExcessRet_{f,3Y} - ExpenseRatio_f
$$

$$
NetAlphaScore_f = Score(NetExcessRet_{f,3Y})
$$

13.3 CostEfficiencyScore

$$
CostEfficiencyScore_f =

$$
0.6 \cdot AlphaPerCostScore_f
+0.4 \cdot NetAlphaScore_f
$$

***

14. LAYER 10 – INVESTOR DECISION LAYER

Goal: convert quantitative picture to human-usable attributes.

14.1 Role: Core vs Satellite

$$
CoreLikelihood_f =
0.4 \cdot FundamentalQuality_f
+0.2 \cdot ManagerSkill_f
+0.2 \cdot RegimeRobustness_f
+0.2 \cdot RiskBehaviorScore_f
$$

Rule:

- If $CoreLikelihood_f \geq 0.7$ and $Category_f \in$ {Large Cap, Flexi Cap, Large & Mid, Aggressive Hybrid} $\rightarrow$ $Role_f$ = CORE.
- Else $\rightarrow$ $Role_f$ = SATELLITE.

14.2 SIP vs Lump Sum suitability

$$
SIPPreference_f =
0.5 \cdot RiskBehaviorScore_f
+0.5 \cdot TimeStructureScore_f
$$

If $SIPPreference_f \geq 0.6$ $\rightarrow$ $SIPSuitable_f$ = TRUE.

LumpSumPreference_f can be flagged TRUE if:

- CrashScore_f ≥ 0.6 and RiskBehaviorScore_f ≥ 0.6.

14.3 Risk bucket

Raw riskiness:

\[
RiskScoreRaw_f = 1 - RiskBehaviorScore_f
\]

Buckets:

- Conservative: RiskScoreRaw_f < 0.3
- Moderate: 0.3 ≤ RiskScoreRaw_f < 0.6
- Aggressive: ≥ 0.6

14.4 Horizon compatibility

Based on Category_f and risk bucket:

- Equity large/flexi: recommended horizon ≥ 5 years (Long).
- Small/Mid: ≥ 7 years (Very Long).
- Hybrid: ≥ 3–5 years (Medium/Long).

Output: HorizonBucket_f ∈ {Short, Medium, Long, VeryLong}.

***

15. LAYER 11 – ML FEATURE OUTPUT LAYER

Expose:

1. **Raw primitives**: all metrics used above (returns, alphas, vol, CVaR, flows, exposures, etc.).
2. **Normalized scores**: all layer scores and sub-scores.
3. **Time series**: monthly alpha, excess returns, flows, style exposures.
4. **Regime labels**: time-stamped.

This layer is just a structured feature dictionary for ML; no extra math.

***

16. LAYER 12 – FINAL DECISION ENGINE

16.1 Inputs

Top scores:

- $Q_f = FundamentalQuality_f$
- $S_f = ManagerSkill_f$
- $F_f = ForwardOutperformanceProb_f$
- $RReg_f = RegimeRobustness_f$
- $PInt_f = PortfolioQualityScore_f$
- $E_f = EdgeScore_f$
- $Risk_f = RiskBehaviorScore_f$
- $Cost_f = CostEfficiencyScore_f$
- $TS_f = TimeStructureScore_f$

16.2 Interaction terms

1. Quality × Skill:

$$
QS_f = \sqrt{Q_f \cdot S_f}
$$

2. Skill × Forward:

$$
SF_f = \sqrt{S_f \cdot F_f}
$$

3. Edge × Risk penalty:

$$
EdgeRiskPenalty_f =
\begin{cases}
(1 - Risk_f) \cdot E_f, & \text{if } E_f < 0.4 \text{ and } Risk_f < 0.4 \\
0, & \text{otherwise}
```
```

\end{cases}
\]

4. Regime × Portfolio penalty:

\[
RegimePenalty_f = \max(0,\ 0.5 - RReg_f) \cdot (1 - PInt_f)
\]

16.3 Aggregation

Base linear:

\[
Base_f =
 0.22 \cdot Q_f
+0.18 \cdot S_f
+0.15 \cdot F_f
+0.10 \cdot RReg_f
+0.10 \cdot PInt_f
+0.08 \cdot E_f
+0.10 \cdot Risk_f
+0.07 \cdot Cost_f
\]

Non-linear bonus:

\[
Bonus_f = 0.10 \cdot QS_f + 0.10 \cdot SF_f
\]

Penalty:

\[
Penalty_f = 0.15 \cdot EdgeRiskPenalty_f + 0.10 \cdot RegimePenalty_f
\]

Raw decision:

\[
RawDecision_f = Base_f + Bonus_f - Penalty_f
\]

Final DecisionScore:

\[
DecisionScoreMF_f = \min\big(1,\ \max(0,\ RawDecision_f)\big)
\]

16.4 Confidence

Agreement variance across main scores:

\[
VarScores_f = Var(Q_f, S_f, F_f, RReg_f, PInt_f, E_f, Risk_f, Cost_f, TS_f)
\]

\[
AgreementScore_f = Score(-VarScores_f)
\]

Data coverage: fraction of primitives not imputed (0.5). Denote DataCoverage_f ∈.

\[
Confidence_f = 0.7 \cdot AgreementScore_f + 0.3 \cdot DataCoverage_f
\]

16.5 Action mapping and churn control

- If DecisionScoreMF_f > 0.75 → Recommendation_f = STRONG_BUY
- Else if 0.60 < DecisionScoreMF_f ≤ 0.75 → BUY
- Else if 0.40 < DecisionScoreMF_f ≤ 0.60 → HOLD
- Else if 0.25 < DecisionScoreMF_f ≤ 0.40 → TRIM / AVOID_NEW
- Else → SELL / EXIT

Churn control:

- Let previous score and recommendation at T_{prev}: $DecisionPrev_f, RecPrev_f$.
- Only change recommendation if:

$$
|DecisionScoreMF_f - DecisionPrev_f| > 0.10 \quad \text{or} \quad (T - T_{prev}) \ge 30\ \text{days}
$$

Else keep RecPrev_f.

***

17. System Flow (Architecture Summary)

1. **Data ingestion**: NAV, benchmark, portfolios, flows, AUM, macro regimes.
2. **Pre-processing**: clean series, align dates, compute daily and monthly returns.
3. **Feature construction**: all primitive metrics (returns, alphas, vol, CVaR, exposures, flows, etc.).
4. **Scoring layers (1–9)**: compute per-layer scores $[0,1]$.
5. **Investor layer**: derive role, SIP/Lumpsum suitability, risk bucket, horizon.
6. **ML outputs**: expose raw and normalized features, sequences, regimes.
7. **Final decision engine**: combine layers via non-linear aggregation and interaction terms → DecisionScoreMF, Confidence, Recommendation.
8. **API output**: per-fund JSON with DecisionScoreMF, recommendation, factor_breakdown, macro_weights, derived_fields, role, suitability, confidence.